

Chapter 81 — Minimally Invasive RPLND — Numbers, Gold Standards and Traps

IF YOU READ NOTHING ELSE

- **MI-RPLND must duplicate open RPLND** — approach never alters extent.
- **Postchemotherapy standard:** bilateral full template + nerve-sparing.
- **Stage I / small-volume II:** extended unilateral nerve-sparing template acceptable.
- **Boundaries:** renal hila ↑, ureters laterally, ureter–common iliac crossing ↓.
- **Lumbar vessels must be divided** → retroaortic/retrocaval clearance.
- **Conversion** ≈5%; bleeding = unique MI complication.
- **Antegrade ejaculation** ≥90% after primary MI-RPLND.
- **No absolute contraindications** to MI-RPLND.
- **RFS** ≈85% limited pN+ without adjuvant chemo; ≈80% any pN1–3.
- **Always therapeutic intent** — never a staging biopsy.

INDICATIONS AND CONTEXT

- **RPLND (retroperitoneal lymph node dissection), chemo-naïve:** marker-negative stage I NSGCT (nonseminomatous germ cell tumour); small-volume stage II seminoma/NSGCT.
- **Postchemotherapy:** NSGCT residual mass >1 cm; select seminoma.
- **20–30%** with enlarged nodes are **pN0** → chemo avoided.
- **First laparoscopic RPLND 1992; first robotic 2006.**
- **Laparoscopic vs open:** ↓ stay, ↓ blood loss, ↑ operative time.

GUIDELINES

- **NCCN (National Comprehensive Cancer Network):** silent.
- **AUA (American Urological Association):** may offer — GCT + MIS expertise; no long-term data.
- **EAU (European Association of Urology):** specialist centres only, expert in open + MIS.

CONTROVERSY

- **Most controversial operation** in urological oncology; **no level 1 data.**
- **pN0 rates up to 83%** in MI series → selection concern.
- **Stage IA → observe**, except teratoma with **somatic transformation.**
- **No large series >5 years;** atypical relapse incl. **peritoneal carcinomatosis.**

SURGICAL TECHNIQUE

- **Imaging** ≤4 weeks, **markers** ≤1–2 weeks preoperatively.
- **Enlarged nodes intraoperatively** → consider bilateral full template.
- **GETUG T93BP (BEP ×3 vs EP ×4):** relapses **70%**, deaths **50%** in unoperated/incomplete RPLND.
- **Nerve-sparing:** postganglionic fibres **L1–L4** → hypogastric plexus **below IMA** (inferior mesenteric artery).
- **≈3 efferent fibres** per sympathetic chain.

PERIOPERATIVE PREPARATION

- **Safety suture:** 4-0/5-0 Prolene, RB needle, LAPRA-TY ≈4 inches.
- **Prior bleomycin** → review fluids and oxygenation.
- **Open set in theatre;** second time-out.

PATIENT AND TUMOUR FACTORS

- **Relative contraindications:** extensive prior abdominal surgery; morbid obesity.
- **MI candidate stage II:** 1–2 nodes, primary landing zone, **none** >3 cm.

- **Ideal post-chemo MI:** solitary landing-zone mass, small response (teratoma).
- **Open if:** post-chemo, large residual mass, big size change (desmoplasia), vessel resection.
- **MI less suitable:** somatic malignancy; retrocrural disease.

COUNSELLING AND PREPARATION

- **High-fat diet day before** → lymphatics visible.
- **No full bowel prep;** ERAS (Enhanced Recovery After Surgery) + carbohydrate loading.
- **Lumbar control:** Weck clip vessel side, sealer specimen side.

POSITIONING

- **Modified lateral decubitus 45°** — unilateral; ports as renal, more caudal.
- **Laparoscopic supine:** 4 × 12-mm trocars, first **1 cm below xiphoid.**
- **Steep Trendelenburg:** camera **4–6 cm below umbilicus;** bilateral without repositioning.
- **Docking:** Si over left shoulder; Xi from side.

TEMPLATE DISSECTION

- **Right:** Toldt → foramen of Winslow; Kocherise duodenum.
- **Right hazard:** right renal artery in interaortocaval space; accessory lower-pole arteries.
- **Left:** medialise spleen + pancreatic tail; gonadal vein divided at left renal vein.
- **Left renal vein** = lymphatic hotspot → clip liberally.
- **Fascial closure:** extraction site + ports ≥10 mm; **no routine drain.**
- **Bag packets** (Endo Catch) immediately.

ADVERSE EVENTS

- **Bleed:** pressure + ↑ pneumoperitoneum → prepare → undock, convert.
- **Blood on lens** → single-eye setting or move camera.
- **No blind clipping or suturing.**
- **Lymph leaks** may hide even at low pressure.

POSTOPERATIVE AND OUTCOMES

- **Discharge 1–2 days.**
- **Chylous ascites** ≈ open rates.
- **Open node-yield benchmark** ≈30.
- **Post-chemo MI data:** small series — no conclusions.
- **miR-371a-3p** → future selection for RPLND and adjuvant chemo.

CLASSIC TRAPS

- **Trap:** conversion = complication — **Actually:** may be elective, for safety or oncology.
- **Trap:** low MI recurrence proves equivalence — **Actually:** confounded by pN0 + adjuvant chemo.
- **Trap:** pIIA after RPLND → adjuvant chemo — **Actually:** observation preferred.
- **Trap:** stage IA → RPLND — **Actually:** observe (except somatic transformation).
- **Trap:** MI allows smaller template — **Actually:** extent identical to open.
- **Trap:** sealers secure lymphatics — **Actually:** unreliable; clip.
- **Trap:** MI contraindicated post-chemo — **Actually:** no absolute contraindication; open favoured.
- **Trap:** guidelines endorse MI — **Actually:** NCCN silent; AUA/EAU conditional.
- **Trap:** lowering pressure excludes leak — **Actually:** low-pressure lymphatics may still hide.
- **Trap:** laparoscopic staging RPLND adequate — **Actually:** no lumbar division → dual therapy.